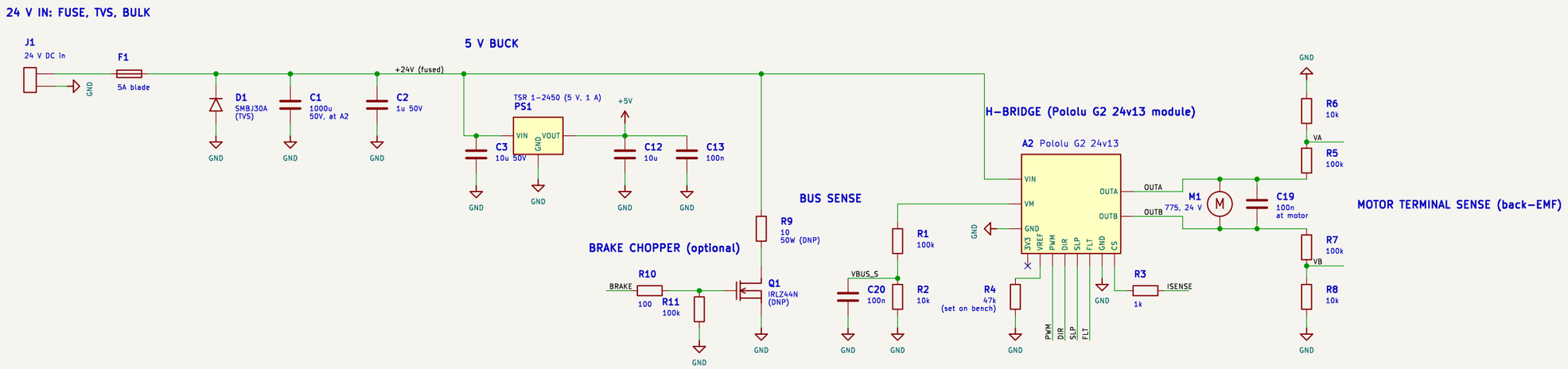


NOTES. A concept drawn by Claude at Bob Drzyzgula's direction, October 2026: not built, not tested, values not checked.

1. The mains version's control side unchanged; the triac, zero-cross, relay and isolated supply replaced by a 24 V adapter, a buck and an H-bridge.
2. Pins (RP2350B): PWM GP4 (20 kHz, PWM slice 2); DIR GP5; SLP GP6; FLT GP13 (internal pull-up); BRAKE GP1; HALL GP7; ENC_A/B GP2/GP3; STITCH GP10; HEEL GP11; NEEDLE GP12; knee, pedal, FSR, limit ADC0-3; ISENSE ADC4 (GP44); VA, VB ADC5, ADC6; VBUS_S ADC7 (GP47).
3. A2 truth table: PWM high drives (DIR sets the way); PWM low brakes (both outputs low). Its PWM input is pulled low on the board, so a resetting MCU leaves the motor braked. CS is about 40 mV/A, valid only while driving: sample it mid on-time. No capacitor on CS.
4. R4 lowers A2's current limit from its default ~30 A (100k gives ~16 A); aim for 8-10 A and set it on the bench with the motor stalled.
5. Slow down by braking (PWM low), which spends the energy in the motor's own resistance. Never reverse at speed: that pumps the bus up. Watch VBUS_S: if it rises past ~28 V, stop driving and fire the brake chopper (Q1, R9), if fitted. D1 only catches short spikes.
6. VA, VB read each motor terminal (/11): their difference is the motor's voltage; with A2 asleep (SLP low) it is the back-EMF, the speed.
7. C19 and two 47n caps from each terminal to the motor case go at the motor; twist the motor leads. PS1 feeds VSYS through D2, as before.

Drafted by Claude at the direction of Robert Drzyzgula; not built or tested
The mains controller's control side, driving a 775 brushed PM motor through a G2 H-bridge

Sheet: /
File: mcu_controller_dc.kicad_sch

Title: VSM MCU controller, 24 V DC motor variant (concept)

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